

# 《门前惊雷》DLC 兼容与适配指南

## （双语版本）

汉化：霜泽图书馆汉化组

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### 霜泽图书馆适配建议

- 1、老生常谈的 **state**，需要覆盖，**nudge** 分战略区和 **building**，有 30 个
- 2、覆盖你 MOD 里面的 **Define**，否则很容易出现 **ai** 报错闪退（原理是 **ai** 瞎操作，导致误触了没适配的定义内容，闪退）
- 3、动态 **icon** 的代码写法大改，原代码失效，需要及时更改
- 4、老生常谈的陆军，海军编制啥的需要适配一下（注意：原先在 **unit** 里面的 **add\_temporary\_buff\_to\_unit** 代码块失效，文档里面写明要写在 **unit\_modifier** 内）
- 5、**Bookmark** 现在新建了一些分类方式，但是 **P** 社尚未提供接口修改（很逆天）
- 6、科技、编制、选择国家页面的 **UI** 需要适配，否则容易闪退。

霜泽图书馆群号：378525932，如有错误，欢迎提出

# TAOG 兼容与适配指南



我对加入团级支援连的真实反应 ([图片来源](#))

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模组制作者们，是时候迎接另一个重大更新了！这次，部队设计及其配套的科技树是主角。

我通常将新功能分为玩家功能（玩家会直接与之互动的功能）和模组制作者功能（玩家不会直接互动的功能）

## 关于 DLC 的说明

每个 DLC 都包含免费和付费功能，TAOG 也不例外。

我不认为带有 DLC 元素的功能（比如团级支援）是 DLC 功能。它们只是有一些 DLC 锁定的元素。只有当整个功能被锁定时，该功能才被视为 DLC 功能。

# 新支援连：主要的玩家功能

支援连已经完全重做。现在分为两类：**师级**和**团级**。

• **师级支援连**是你熟悉的一类：它们位于侧面，包括工兵连、野战医院等。通常，师级单位会提供大量增益，如堑壕值、伤兵复原、可靠性以及大量的软攻或硬攻。这也是你能找到大多数百分比修正以及营级倍率的地方！

• **团级支援连**位于基础部队的下方，在它们上方至少需要有三个（mod 可自定义）基础单位，并且通常给予较小的加成。它们包括野战炮、重武器（如机枪和迫击炮）、防空和反坦克武器等。有趣的是，坦歼和自行防空炮也有支援连版本。此外，它们可以要求上方有特定类型的基础营。例如：

```
allowed_battalion_groups = {infantry mobile combat_support }
```

这些被简单地归类为

```
regimental = no  
  
divisional = no
```

请注意，这两个参数都是可选的。你可以设置一个单位为师级、团级，或两者兼有！

如果你有一个新的支援连，不确定该设为哪种，以下清单可以帮助你判断：

- 它是否提供百分比加成，例如伤兵复原、恢复速度或营级倍率？
- 它的属性/花费是否与师属支援火炮、防空炮或反坦克炮相似？
- 每个师一个是否够用？

如果你对以上任何一条回答"是"，则将其设为**师级**。否则，将其设为**团级**。

# 数学表达式：主要的 modder 功能

数学表达式允许使用常量、作用域内的变量和数学函数进行数学计算。使用数学表达式预计比使用临时变量进行计算性能更高。

所有数学表达式都基于定点算术。对于布尔运算，0.0 被视为假（false），任何其他值都被视为真（true）。所有返回布尔值的运算符在真时返回 1.0，假时返回 0.0。

数学函数列表可在 [TAOG 兼容与功能指南](#) 找到。

## 语法

数学表达式使用一个基础值（“value = ...”），然后由顺序执行的语句进行修改。

语句可以是：

- **无参数语句**（例如 round = yes），只是对累加器应用一项操作。
- **简单语句**，以一个或多个数学表达式作为参数（例如 add = 5, clamp = { min = 0 max = 100 }）。
- **控制流语句**，以一个语句块作为参数（例如 if = { limit = { value = x greater\_than = 10 } add = 100 }）。
- **集合迭代器**，以一个命名集合和一个修改语句块作为参数，对每个元素应用操作。将作用域切换至集合内的元素。

## 变量

除了 modulo\_variable 和 clamp\_variable 外，所有临时变量和普通变量的效果都支持数学表达式

## 错误处理

如果数学表达式未能正确解析，运行时的操作数值将被替换为 0.0。

## 示例

```
***
{ value = num_cavalry
  add = num_motorized
  add = num_mechanized
  greater_than = { value = num_units 乘以 = 0.4 }
}
***
```

## 数学函数列表

### 乘法

将累加器乘以一个值。

```
### 示例
...

{ value = 2.3 multiply = num_units }

...
```

### 加法

它会给累加器增加一个值。

```
### 示例
...

{ value = 5.0 add = num_factories }

...
```

### 减法

从累加器中减去一个值。

```
### 示例
...

{ value = 20 subtract = 5 }

...
```

### 除法

将累加器除以一个值。

```
### 示例
```

```
...
```

```
{ value = 100 divide = 5 }
```

```
...
```

### 取最小值

将累加器设为它自身与给定值中的最小值。**注：**在这个例子中，这意味着结果将是 4

```
### 示例
```

```
...
```

```
{ value = 9 min = 4 }
```

```
...
```

### 取最大值

将累加器设为它自身与给定值中的最大值。**注意：**在这个例子中，这意味着结果将是 15

```
### 示例
```

```
...
```

```
{ value = 9 max = 15 }
```

```
“”
```

### clamp

将累加器固定在最小值和最大值之间。**注意**参数有**顺序**要求。

```
### 示例
```

```
...
```

```
{ value = num_units clamp = { min = 0 max = 100 } }
```

```
'''
```

>

如果累加器大于该值，则返回 1，否则返回 0。

### 示例

```
'''
```

```
{ value = 10 greater_than = 5 }
```

```
'''
```

<

如果累加器小于该值，则返回 1，否则返回 0。

### 示例

```
'''
```

```
{ value = 2 less_than = 5 }
```

```
'''
```

>=

如果累加器大于或等于该值，则返回 1，否则返回 0。

### 示例

```
'''
```

```
{ value = 5 greater_than_or_equals = 5 }
```

```
'''
```

<=

如果累加器小于或等于该值，则返回 1，否则返回 0。

```
### 示例
...

{ value = 5 less_than_or_equals = 5 }

...
```

## 等于

如果累加器等于给定值，则返回 1，否则返回 0。

```
### 示例
...

{ value = 7 equals = 7 }

“”
```

## 不等于

如果累加器不等于给定值，则返回 1，否则返回 0。

```
### 示例
...

{ value = 7 not_equals = 5 }

...
```

## 取整

将累加器四舍五入到最近的整数。

```
### 示例
...
```



```
{ value = 3.7 round = yes }  
...
```

基于条件修改累加器的条件语句。

请注意，`limit` 是一个数学表达式。如果值不为 0，则被视为 `true`。

```
### 示例  
...  
  
{ value = x  
  if = { limit = { value = x greater_than = 10 } add = 100 }  
  else = { subtract = 1 } }  
...
```

## 遍历集合

对已命名集合进行遍历，并对集合中的每个元素应用语句。

```
### 示例  
...  
  
{ value = 0  
  every_collection = {  
    named_collection = my_collection  
    add = element_value } }  
...
```

## 陆军总部： DLC 玩家功能

你现在可以将将领部署到师一级以获得额外的加成，但将领有被俘的风险。这需要一些调整并加入了新功能。如果你有不想被俘的将领（比如国家领导人），你应该添加参数：

```
can_be_captured = no
```

代码在角色文件中 **name** 代码块下面。

单位现在支持了一些新的语句：

```
deployed_leader_modifiers = {  
    river_crossing_factor = 0.2  
}  
  
allow_in_army_hq = yes  
  
allow_in_non_army_hq = no  
  
required_dlc = { “Thunder at Our Gates” }  
  
    deployment_cost = 1  
  
    enable_ability = { deeper_dig_in }
```

在 `history/general/` 文件夹中，你需要添加一个新文件：`taog_hq_template.txt`。这会设定一个通用的总部模板，还有其他你需要的独特模板！**你必须为游戏中每个国家准备一个总部模板，否则游戏将崩溃！！**

## 舰长：DLC 玩家功能

另一个主要功能是舰长，类似于师级指挥官。相应地，勋章也做了些微调。例如，过去的代码：

```
unit_modifiers = {  
    army_org_factor = 0.05  
    army_strength_factor = 0.05  
}  
  
one_time_effect = {  
    add_divisional_commander_xp = 100  
}
```

现在的代码：

```
unit_modifiers = {  
    army_org_factor = 0.05  
    army_strength_factor = 0.05  
}  
  
ship_modifiers = {  
    navy_org_factor = 0.05  
    naval_critical_effect_factor = 0.05  
}  
  
officer_xp = 100
```

Officer\_xp 是用于经验值的新效果！

一般来说，以下效果在原版中被用作等效项，除了一些符号变化（以保持正面增益）：

| 陆军效果                      | 海军效果                               |
|---------------------------|------------------------------------|
| army_morale_factor        | naval_morale_factor                |
| experience_loss_factor    | experience_gain_navy_unit_factor   |
| supply_consumption_factor | naval_supply_consumption_factor    |
| recon_factor              | naval_detection                    |
| max_dig_in_factor         | naval_mines_effect_reduction       |
| army_defence_factor       | naval_defense_factor               |
| breakthrough_factor       | naval_critical_score_chance_factor |
| army_attack_factor        | naval_damage_factor                |
| army_org_factor           | navy_org_factor                    |

|                               |                                   |
|-------------------------------|-----------------------------------|
| army_fuel_consumption_factor  | navy_fuel_consumption_factor      |
| equipment_capture_factor      | naval_ship_recovery_chance        |
| recon_factor_while_entrenched | navy_visibility 或 spotting_chance |
| army_strength_factor          | naval_critical_effect_factor      |
| armor_factor                  | critical_receive_chance           |

# 扩展的玩家功能

以下是玩家会关心的更新和新内容，这些不属于全新的类别。

## 新的特种部队学说

特种部队现在使用带有路线的新精通系统。这也带来了一些新的 mod 功能：

```
allow_in_multiple_tracks = yes

# 可选 – 如果是 yes，该子学说可以*同时*分配给多个路线。默认是 no。

xor = { other_subdoctrine_a other_subdoctrine_b }

# 可选 – 列出不能在同一文件夹内的不同路线中同时激活的子学说。在同一路线上替换为互斥的
子学说仍然是允许的。
```

## 全新的关键一击模块

船体和模块都新增了大量的关键一击模块！如果你有独特或扩展的模块，可以相应地扩展它们。

# 扩展的 modder 功能

## 更多 AI 海军相关内容

特遣舰队有一个新的参数：

```
keep_updated = yes
```

这个功能没有文档，但我猜它决定了最新的舰船去向。

## 国策内嵌进度条

国策内嵌窗口现在支持进度条。

```
scripted_progressbars = { # 应基于变量填充的进度条列表

    icon_name = { # 进度条名称（必须是一个图标，作为 gui_component_name 的子组件或按钮，
其 spriteType 为 progressbartype ）

        progress = <variable_name>
    }
}
```

## 新增突袭选项

现在我们有了海军突击队突袭。以下是一些新的修正项：

```
ai_min_success_chance = 0.1 # [可选] 每种类型对 RAIDS_MIN_SUCCESS_FOR_LAUNCH (0.0-1.0)
的覆盖

max_distance = 500 # [可选] 源到目标路径长度的硬上限，以通用距离单位计（与引擎路径长度相同
空间）。比这更远的源无法使用。当 unlimited_ai_range = yes 时，AI 忽略此限制
```

此外，还有一个全新的自定义成功率修正项部分。请查看文档！

## 重做的技能修正

现在技能的运作方式完全不同了。你熟悉的旧语法：

```
one_time_effect = {  
    add_temporary_buff_to_units = {  
        combat_offense = 0.2  
        combat_breakthrough = 0.25  
        org_damage_multiplier = -1.0  
        str_damage_multiplier = 0.6  
        war_support_reduction_on_damage = 0.2  
        cannot_retreat_while_attacking = 1.0  
  
        days = 7  
        tooltip = ABILITY_FORCE_ATTACK_TOOLTIP  
    }  
}
```

你现在**必须**使用的新语法：

```
unit_modifiers = {  
    attack = 0.2  
    breakthrough_factor = 0.25  
    org_damage_multiplier = -1.0  
    str_damage_multiplier = 0.6  
    war_support_reduction_on_damage = 0.2  
    cannot_retreat_while_attacking = 1.0
```

```
}
```

旧的语法已经不管用了。你必须更改！

此外，还支持了一个新的参数：

```
requires_deployed_hq = yes
```

## 动态国策图标的新语法

你熟悉的旧语法是这样的：

```
icon = {  
    trigger = {  
        has_government = democratic  
    }  
    value = GFX_focus_GER_german_hegemony_in_the_middle_east_demcoratic  
}  
icon = {  
    trigger = {  
        NOT = {  
            has_government = democratic  
        }  
    }  
    value = GFX_focus_GER_german_hegemony_in_the_middle_east_neutrality  
}
```

那已经没了。现在是：

```
icon = {  
    GFX_focus_GER_german_hegemony_in_the_middle_east_demcoratic = {  
        has_government = democratic  
    }  
}
```

```
}  
  
GFX_focus_GER_german_hegemony_in_the_middle_east_neutrality = yes  
  
}
```

旧的语法已经不管用了。你必须更改！

## 同步动态令牌

某些动态令牌（如自定义 `modifier_definitions`）存在一定限制。这可能会在多人游戏中引发 OOS（不同步）问题。现在这些令牌大多数会在错误日志中给出错误。

错误如下。

```
令牌<token_name>是动态令牌，这可能会根据使用方式导致 OOS，请将其添加为同步动态令牌以防止 OOS
```

这些令牌现在可以添加到 `common/synchronized_dynamic_tokens` 的文本文件中，以消除它们导致 OOS 的风险。

## 新效果

### `release_from_captivity`

角色作用域

将作用域内的陆军指挥官从囚禁中释放。如果该指挥官未被俘，则无效。

```
### 示例  
  
var: target_leader = { release_from_captivity = yes }
```

### `unlock_subunit`



国家作用域

为该国家解锁指定的子单位。

### 示例

```
GER = {  
    unlock_subunit = rangers_support  
}
```

**release\_captured\_generals\_from**

国家作用域

使作用域内的国家释放关押在指定省份的、来自指定国家的所有被俘将军。

如果未指定省份，则释放所有将军。

### 示例

```
release_captured_generals_from = {  
    target = GER  
    province = <province> ( optional )  
}  
  
release_captured_generals_from = FROM
```

**capture\_by**

使作用域内的部队指挥官被指定国家俘获（需要是将军）。

可选定关押该指挥官的省份。

### 示例

```
capture_by = GER
```

```
capture_by = FROM
```

```
capture_by = {  
    country = GER  
    province = 1234  
}
```

## 新触发器

### **can\_be\_captured**

角色作用域

**can\_be\_captured** = yes/no —— 检查当前角色在部署到师后是否可以被俘获。

### **is\_leader\_visible**

国家作用域

检查领袖是否对其拥有者可见（领袖脚本中的 **is\_visible** 触发器）

### **is\_in\_state**

角色作用域

检查角色是否处于特定地区。

适用于部署的将军（指挥部位置）、科学家（设施位置，包括阵营项目）和特工（任务位置）。

### 示例

# 检查将军是否处于特定地区

```

any_unit_leader = {
    is_in_state = 64 # 勃兰登堡
}

# 查看首都是否有科学家
every_character = {
    limit = { is_active_scientist = yes }
    is_in_state = root.capital
}

```

### has\_completed\_goal

国家作用域

检查该国在完成特定目标时是否为某阵营的成员。

与`has\_completed\_faction\_goal`不同，此触发器在阵营解散/重建后仍然保持。

```

### 示例

TAG = {
    has_completed_goal = goal_id
}

```

### debug\_math\_expr

任何作用域

计算一个数学表达式并将数值结果写入提示框。

始终返回 true，用于调试脚本。

```

### 参数

- 表达式：任何支持的数学表达式。

```

### 示例

```
debug_math_expr = { value = 120 add = 15 clamp = { min = 0 max = 100 } }
```

### **has\_any\_general\_captured\_by**

检查作用域内的国家是否有任何部队指挥官被目标国家俘获。

### 示例

```
FRA = {  
    has_any_general_captured_by = GER # 如果法国的任何将军被德国俘虏，将返回 true  
}
```

### **is\_general\_captured**

角色作用域

检查作用域内的部队指挥官（将军）是否被（任何国家）俘获。

### 示例

```
my_character = {  
    is_general_captured = yes  
}
```

### **has\_any\_captured\_general**

国家作用域

检查作用域内国家的任何将军是否被另一个国家俘获。

### 示例

```
FRA = {  
    thas_any_captured_general = yes  
}
```

### **is\_promoted\_from\_unit**

角色作用域

检查部队指挥官是否是从单位（师或舰船）晋升而来的。

#### ### 示例

```
any_unit_leader = {  
    is_promoted_from_unit = yes  
}
```

### **check\_expr**

任何作用域

计算一个数学表达式，当结果非零时返回 true。

#### ### 参数

– 表达式：任何支持的数学表达式。

#### ### 示例

```
check_expr = { value = 2 add = 3 greater_than = 4 }
```

## 新修正

| 修正 (Modifier)                   |
|---------------------------------|
| army_non_core_attack_factor     |
| cannot_retreat_while_attacking  |
| army_non_core_defence_factor    |
| cannot_retreat_while_defending  |
| casualty_trickleback            |
| combat_entrenchment             |
| org_damage_multiplier           |
| str_damage_multiplier           |
| war_support_reduction_on_damage |

## 新变量

| 变量 (Variable)               | 说明 (Description)    |
|-----------------------------|---------------------|
| days_until_next_election    | 距离下次选举的天数。如果没有则为 -1 |
| convoys_destroyed           | 该国摧毁的运输船总数          |
| avg_unit_entrenchment_ratio | 所有单位的平均堑壕比率         |

## 新 Define 定义

1.

CAPTAIN\_EXPERIENCE\_ON\_SHIP\_MULT = 1.55,
2.

—— 舰船获得经验乘以此值，应用于舰长

- 1. MAX\_CAPTAIN\_EXPERIENCE\_ON\_SHIP = 8000,
- 2. — 舰长可获得的最大经验值

- 1. CAPTAIN\_EXPERIENCE\_ON\_SHIP\_PER\_MEDAL\_MULT = 0.1,
- 2. — 舰长经验获得乘以此值\*应用时的舰船勋章数量

- 1. MAX\_REGIMENTAL\_SUPPORT\_WIDTH = 5,
- 2. — 师设计中团级支援的最大宽度
- 3. MAX\_REGIMENTAL\_SUPPORT\_HEIGHT = 1,
- 4. — 师设计中团级支援的最大高度
- 5. MAX\_HQ\_BATTALION\_WIDTH = 1,
- 6. — 师设计中（陆军指挥部模板）团的最大宽度
- 7. MAX\_HQ\_BATTALION\_HEIGHT = 4,
- 8. — 师设计中（陆军指挥部模板）团的最大高度
- 9. MAX\_HQ\_SUPPORT\_WIDTH = 1,
- 10. — 师设计中（陆军指挥部模板）支援的最大宽度
- 11. MAX\_HQ\_SUPPORT\_HEIGHT = 4,
- 12. — 师设计中（陆军指挥部模板）支援的最大高度
- 13. MAX\_HQ\_REGIMENTAL\_SUPPORT\_WIDTH = 0,
- 14. — 师设计中（陆军指挥部模板）团级支援的最大宽度
- 15. MAX\_HQ\_REGIMENTAL\_SUPPORT\_HEIGHT = 0,
- 16. — 师设计中（陆军指挥部模板）团级支援的最大高度

- 1. REGIMENTAL\_SUPPORT\_REQUIRED\_BATTALIONS = { 3 },
- 2. — 对于每个团级支援行，团中需要多少个营才能在该行放置支援。

- 1. AI\_BATTALION\_BUILD\_ORDER = { 1, 6, 11, 16, 21,
- 2. 2, 7, 12, 17, 22,
- 3. 3, 8, 13, 18, 23,
- 4. 4, 9, 14, 19, 24,
- 5. 5, 10, 15, 20, 25 },

6. — 当 AI 决定在哪里放置营时，它会尝试根据此网格放置在编号最低的位置。

1. **REGIMENTAL\_SUPPORT\_SLOT\_COST\_MULTIPLIER = 0.5,**
2. — 团级支援槽位成本相对于普通支援槽位按此值缩放

1. **UNIT\_LEADER\_MODIFIER\_COOLDOWN\_ON\_DEPLOY = 3,**
2. — 单位指挥官部署的基础冷却天数（未部署时）。为 0 时瞬间完成。按指挥部模板人力缩放。
3. **UNIT\_LEADER\_MODIFIER\_COOLDOWN\_ON\_DEPLOY\_MIN = 1,**
4. — 部署单位指挥官的最短冷却天数，即使对于非常小的指挥部模板也是如此
5. **UNIT\_LEADER\_MODIFIER\_COOLDOWN\_ON\_REDEPLOY = 15,**
6. — 单位指挥官重新部署到新单位的天数。为 0 时瞬间完成。
7. **UNIT\_LEADER\_MODIFIER\_COOLDOWN\_ON\_WITHDRAW = 3,**
8. — 撤回已部署单位指挥官的基础冷却天数。为 0 时瞬间完成。按指挥部模板人力缩放。
9. **UNIT\_LEADER\_MODIFIER\_COOLDOWN\_ON\_WITHDRAW\_MIN = 1,**
10. — 撤回单位指挥官的最短冷却天数，即使对于非常小的指挥部模板也是如此
11. **UNIT\_LEADER\_MODIFIER\_COOLDOWN\_REFERENCE\_MANPOWER = 9600,**
12. — 部署/撤回冷却缩放的参考人力值。在此值时，延迟等于基础值。
13. **UNIT\_LEADER\_MODIFIER\_COOLDOWN\_MANPOWER\_EXPONENT = 1.5,**
14. — 人力缩放的指数（>1 代表 大编制师受到更严厉的惩罚）

1. **FIGHTING\_STRENGTH\_DEATH\_THRESHOLD = 0.001,**
2. — 战斗力低于此值时师会因损耗而消失
3. **FIGHTING\_STRENGTH\_HQ\_ALERT\_THRESHOLD = 0.2,**
4. — 战斗力低于此值时已部署指挥官的肖像会有警告发光

1. **COMMANDER\_ABILITY\_BASE\_RANGE = 20,**
2. — 指挥官能力的基础半径范围

1. **COMMS\_MAX\_DISTANCE = 4,**
2. — 如果  $N \geq$  以下数组的大小，最后一个值将被视为重复



1. `PLANNING_CAP_COMMS_SCALING = { 1.0, 0.95, 0.9, 0.85, 0.8 }`,
2.     — 索引 J 处的值是指挥部位于前线后方 J 个省份时对计划上限应用的缩放
3. `PLANNING_CAP_NO_HQ_SCALING = 0.8`,
4.     — 没有指挥部时（无指挥官，或指挥官未部署，或不是相同的根命令）对计划上限应用的缩放
5. `PLANNING_SPEED_COMMS_SCALING = { 1.0, 0.95, 0.9, 0.85, 0.8 }`,
6.     — 同 `PLANNING_CAP_COMMS_SCALING` 但用于计划速度
7. `PLANNING_SPEED_NO_HQ_SCALING = 0.8`,
8.     — 同 `PLANNING_CAP_NO_HQ_SCALING` 但用于计划速度
9. `LEADER_MOD_COMMS_SCALING = { 1.06, 1.04, 1.02, 1.01, 1.0 }`,
10.    — 同 `PLANNING_CAP_COMMS_SCALING` 但用于指挥官修正
11. `LEADER_MOD_NO_HQ_SCALING = 1.0`,
12.    — 同 `PLANNING_CAP_NO_HQ_SCALING` 但用于指挥官修正
13. `ABILITY_COMMS_SCALING = { 1.06, 1.04, 1.02, 1.01, 1.0 }`,
14.    — 同 `PLANNING_CAP_COMMS_SCALING` 但用于主动技能
15. `ABILITY_NO_HQ_SCALING = 1.0`,
16.    — 同 `PLANNING_CAP_NO_HQ_SCALING` 但用于主动技能

1. `GENERAL_PROXIMITY_CLOSE = 1`,
2.     — 在"近"距离设置下，将军应保持在前线后方这么多省份
3. `GENERAL_PROXIMITY_MEDIUM = 2`,
4.     — 在"中"距离设置下...
5. `GENERAL_PROXIMITY_FAR = 3`,
6.     — 在"远"距离设置下...
7. `GENERAL_PROXIMITY_DEFAULT = 1`,
8.     — 已部署将军的默认距离设置。
9. `GENERAL_RANK_TO_ARMY_HQ_EXP_LEVEL_FACTOR = 0.5`,
10.    — 将军的军衔乘以此因子（向上取整）来决定生成的陆军指挥部师的经验等级

1. `ARMY_HQ_REQUISITION_MINIMUM_REMAINING_PERCENTAGE = 10`,
2.     — 部署陆军指挥部时，各师的装备或人力不会被征用到低于目标人力或装备的此百分比

1. `PREFERRED_PRISON_VP = 5`,
2.     — 俘获将军时，尝试寻找至少有这么多胜利点的省份来关押他们。

1. CAPITAL\_SHIP\_COMBAT\_RETREAT\_MULT = 0.5,
2. — 主力舰和航母的战斗撤退阈值乘数（它们能承受更多打击）

1. NAVY\_REPAIR\_BASE\_SEARCH\_NON\_OPERATIONAL\_STR = 0.65,
2. — AI 认定舰队失去作战能力的强度系数（小于等于此值时取消任务并送去修理）

1. NAVAL\_COMBAT\_MEDAL\_CHANCE = 24,
2. — 舰船参加战斗后获得勋章的几率为  $1/N$
3. NAVAL\_COMBAT\_MEDAL\_MIN\_DURATION = 48,
4. — 获得勋章所需的最少战斗小时数
5. NAVAL\_COMBAT\_MEDAL\_LAST\_MEDAL\_LIMIT = 30,
6. — 舰船获得新勋章前的最少间隔天数
7. NAVAL\_COMBAT\_MEDAL\_ALLOW\_CONVOY = false,
8. — 对运输船的海军攻击是否计入勋章

1. AI\_TASKFORCE\_REQUIRED\_RESERVE\_RATIO = 0.2,
2. — 为增援保留在预备队中的所需特遣舰队最优组成的比例（每种类型向上取整）

1. LENDLEASE\_CONVOY\_OVERCOMMIT\_PENALTY\_INTERVAL = 3,
2. — 接收方每超出容量  $N$  艘运输船（包括本次租借），AI 接受分数  $-1$

1. STOP\_TRAINING\_ACTIVE\_COMBAT\_RATIO = 0.10,
2. — 当超过此比例的师处于活跃战斗时，AI 停止所有训练（转而增援）

1. **CONVOY\_DEFICIT\_BUILD\_BOOST\_MAX = 150,**
2. — 对因运输船短缺而增加到运输船建造目标的加成的上限。

1. **AI\_SHIP\_SWAP\_MIN\_DAMAGED\_SHIPS = 2,**
2. — 特遣舰队中受损舰船的最小数量, AI 在此之前不考虑将其换到预备队
3. **AI\_SHIP\_SWAP\_DAMAGE\_THRESHOLD = 0.33,**
4. — 每艘舰船的强度阈值, 低于此值时 AI 认为主力舰/航母受损严重需要换到预备队
5. **AI\_REPAIR\_CANCEL\_MIN\_STRENGTH = 0.75,**
6. — 非预备特遣舰队达到此强度时, AI 将使其停止修理并返回任务

1. **AI\_SURFACE\_COMBAT\_FUEL\_RATIO = 0.70,**
2. — 海军燃料中预留给水面作战的比例 (巡逻、打击部队、制海权)
3. **AI\_CONVOY\_DEFENSE\_FUEL\_RATIO = 0.15,**
4. — 海军燃料中预留给护航的比例 (剩余用于破交)

1. **INVASION\_UNITS\_READY\_AT\_MIN\_PLAN = 0.75,**
2. — 计划值处于最低阈值时所需的单位准备比例
3. **INVASION\_UNITS\_READY\_AT\_MAX\_PLAN = 0.25,**
4. — 计划值非常高 (1.0+) 时所需的单位准备比例

1. **AI\_NAVAL\_GOALS\_UPDATE\_FREQUENCY\_DAYS = 7;**

2. — 每多少天重新生成海军 AI 目标（影响性能）

1. ARMY\_LEADER\_ASSIGN\_WITHDRAW\_DAYS = 7,
2. — 将军在没有指挥部可影响的命令（前线或海军入侵）的这么多天后将从指挥部撤回
3. ARMY\_LEADER\_MIN\_DIVISIONS\_FOR\_HQ = 10,
4. — 陆军必须包含至少这么多师，AI 才会为其部署指挥部单位
5. MAX\_DEPLOYED\_ARMY\_HQS = 5,
6. — AI 不会同时部署超过这么多个陆军指挥部
7. MAX\_CAPTURED\_GENERALS\_TO\_STOP\_HQ\_DEPLOY = 3,
8. — 一旦有这么多陆军指挥官被俘，AI 将停止部署新的陆军指挥部

## Bookmark 标签

作者：Ommissiah

bookmark 标签是最近添加的 Bookmark 筛选系统的一部分。简单来说，标签是一组非预定义的自定义筛选器，你可以将其添加到任何地方。

要使用它们，请打开你的 bookmark 文件并这样做：

```
1. bookmarks = {
2.
3.     bookmark = {
4.
5.         name = "AFTERMATH_NAME"
6.
7.         desc = "AFTERMATH_DESC"
8.
9.         date = 1007.1.1.12
10.
11.        picture = "GFX_bookmark_zebrica"
12.
13.        default_country = "HIP"
```

```
14.
15.         default = yes
16.
17.
18.         filters = { label } ### If you use the vanilla ones, just put it after them
19.                                ### 如果你使用原版筛选器，只需将它们放在后面
```

就是这样。你的 **bookmark** 现在支持标签了，如果你添加了任何标签，它们将显示在原版标签之后。如何添加？ 以下是一个示例：

```
1.  "TAG1"={
2.
3.     history = ""
4.
5.     ideology = ...
6.
7.     label = { label_1 label_2 }
8.
9.     ideas = {
10.
11.
12.
13.     }
14.
15.     focuses = {
16.
17.     }
18.
19. }
20.
21. "TAG2"={
22.
23.     history = ""
24.
25.     ideology = ...
26.
27.     label = { label_1 label_3 }
28.
29.     ideas = {
30.
31.     }
32.
```

```

33.     focuses = {
34.
35.     }
36.
37. }

```

这将相应地添加标签 `label\_1`、`label\_2` 和 `label\_3`。如果你选择 `label\_1`，两个国家都会显示。选择其余任何一个标签则只会相应地显示 TAG1 或 TAG2。

你应该知道的一些事情：

|                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1. 使用原版筛选器并非强制要求。从筛选器列表中同时移除 `new_content` 和 `continent` 是完全可行的（我建议你这样做）。                                                                   |
| 2. 国家可以拥有的标签数量没有硬性限制。                                                                                                                       |
| 3. 你不需要在任何地方预定义它们。基本上是"在需要的地方添加"即可。如果书签中至少有一个国家使用了该标签，该标签就会被添加到筛选器列表中。                                                                      |
| 4. 你无法定义这些标签的顺序。我不确定顺序是如何选择的，但似乎是每次游戏重新加载或标签列表更新时，顺序会被重新随机化。                                                                                |
| 5. 其背后的 GUI 是反直觉的，且你能做的调整不多。列表宽度会使所有元素（例如标签）均匀分布以覆盖该宽度。                                                                                     |
| 要更改筛选器的宽度，你需要修改 <code>frontendgamesetupview.gui</code> 文件中的筛选器 <code>OverlappingElementsBoxType</code> 。                                    |
| 6. 标签的本地化键就是你用来定义它的同一个键。所以如果你的国家有 <code>label_1</code> ，你就将其添加为本地化键。它们只能是一行文本，你只能通过调整对应按钮的宽度（同一文件中的 <code>country_filter</code> 容器）来更改文本宽度。 |
| 7. 筛选器没有提示框，然而为了添加类似的功能，你可以利用它们单行文本的特点。                                                                                                     |
| 添加一个像这样的本地化键：                                                                                                                               |
| "共产主义\n\n此国要么以共产主义作为其主要意识形态开局，要么可以将其作为故事线的一部分切换到该意识形态"                                                                                      |
| 这将使 \n 标记之前的所有内容显示在按钮上，之后的所有内容在你悬停时显示。                                                                                                      |
| 8. 遗憾的是，你无法制作垂直筛选器列表，这很可惜。你也不能添加第二行。                                                                                                        |

# TAOG Compatibility and Feature Guide



*My honest reaction to regimental support companies being added ([image source](#))*

---

It's time for another major update, modders! This time the division designer and its supporting tech tree are the stars.

I have arbitrarily categorized the new features as **player features** (those the player will directly interact with) or **modder features** (those they will not).

## A note on DLC

Every DLC brings both free features and paid features, and TAOG is no different.

I do *not* consider features that have DLC elements, like the regimental supports, to be DLC features. They merely have some DLC-locked elements. Only if the entire feature is locked is the feature considered a DLC feature.

# New Support Companies: Major Player Feature

Support companies have been overhauled completely. There are now two categories: **divisional** and **regimental**.

Divisional are what you're familiar with: they sit off to the side and include stuff like engineers and hospitals. Typically, divisional units give large buffs such as entrenchment, trickleback, reliability, or large piles of soft or hard attack. This is also where you find most percentual modifiers and battalion mults!

Regimental sit below the line units, require at least **three** (define-moddable) line units above them, and typically give smaller bonuses. They include such things as field guns, heavy weapons like machine guns and mortars, anti-air and anti-tank, and more. Interestingly, there are also support versions of tank destroyers and SPAA. Additionally, they can require specific *kinds* of line battalions above them. For example:

```
allowed_battalion_groups = {infantry mobile combat_support }
```

These are classed simply with

```
regimental = no  
divisional = no
```

Note that both arguments are optional. You can have something that is divisional, regimental, or both!

If you have a new support company and are not sure which kind to make it, the following checklist will help:

- Does it give a percentage bonus such as trickleback, recovery, or a battalion multiplier?
- Is it similar stat-wise/per IC to divisional support AA, Artillery, or anti-tank?
- Will one per division be enough?

If you answered “Yes” to any of the above questions, put it as Divisional. Otherwise, put it as Regimental.



# Math Expressions: Major Modder Feature

Math expressions allow for mathematical calculations using constants, scoped variables and mathematical functions. Using math expressions is expected to be more performant than using temporary variables for calculations.

All math expressions are based on fixed point arithmetic. For boolean operations, 0.0 is considered false and any other value is considered true. All operators that return a boolean value will return 1.0 for true and 0.0 for false.

A list of math functions can be found [TAOG Compatibility and Feature Guide](#).

## Syntax

Math expressions use a base value (`value = ...`) that is then modified by sequential statements.

Statements can either be:

- \* No argument statement (e.g. `round = yes`) that simply applies an operation to the accumulator.
- \* Simple statements that takes one or more math expression as argument (e.g. `add = 5`, `clamp = { min = 0 max = 100 }`).
- \* Control flow statements that take a block of statements as argument (e.g. `if = { limit = { value = x greater_than = 10 } add = 100 }`).
- \* Collection iterators that takes a named collection and a block of modification statements to apply for each element. The collection elements will be scoped to.

## Variables

All temp variable and normal variable effects support math expressions except `modulo_variable` and `clamp_variable`

## Error Handling

If a math expression fails to parse properly, it will be replaced with the value 0.0 for runtime operations.

## Examples

```
...
```

```
{ value = num_cavalry  
  
  add = num_motorized  
  
  add = num_mechanized  
  
  greater_than = { value = num_units multiply = 0.4 }  
}  
...
```

## List of math functions

### **multiply**

Multiplies the accumulator by a value.

```
### Example  
...  
  
{ value = 2.3 multiply = num_units }  
...
```

### **add**

Adds a value to the accumulator.

```
### Example  
...  
  
{ value = 5.0 add = num_factories }  
...
```

### **subtract**

Subtracts a value from the accumulator.

### ### Example

\*\*\*

```
{ value = 20 subtract = 5 }
```

\*\*\*

### **divide**

Divides the accumulator by a value.

### ### Example

\*\*\*

```
{ value = 100 divide = 5 }
```

\*\*\*

### **min**

Sets the accumulator to the minimum of itself and the value. **Note:** In this example that means the result will be 4

### ### Example

\*\*\*

```
{ value = 9 min = 4 }
```

\*\*\*

### **max**

Sets the accumulator to the maximum of itself and the value. **Note:** In this example that means the result will be 15

### ### Example

\*\*\*

```
{ value = 9 max = 15 }  
...
```

### **clamp**

Clamps the accumulator between a minimum and maximum bound. Note that arguments are order sensitive.

```
### Example  
...  
  
{ value = num_units clamp = { min = 0 max = 100 } }  
...
```

>

Returns 1 if the accumulator is greater than the value, otherwise 0.

```
### Example  
...  
  
{ value = 10 greater_than = 5 }  
...
```

<

Returns 1 if the accumulator is less than the value, otherwise 0.

```
### Example  
...  
  
{ value = 2 less_than = 5 }  
...
```

### **greater\_than\_or\_equals**

Returns 1 if the accumulator is greater than or equal to the value, otherwise 0.

#### **### Example**

...

```
{ value = 5 greater_than_or_equals = 5 }
```

...

### **less\_than\_or\_equals**

Returns 1 if the accumulator is less than or equal to the value, otherwise 0.

#### **### Example**

...

```
{ value = 5 less_than_or_equals = 5 }
```

...

### **equals**

Returns 1 if the accumulator equals the value, otherwise 0.

#### **### Example**

...

```
{ value = 7 equals = 7 }
```

...

### **not\_equals**

Returns 1 if the accumulator does not equal the value, otherwise 0.

### ### Example

---

```
{ value = 7 not_equals = 5 }
```

---

## round

Rounds the accumulator to the nearest integer.

### ### Example

---

```
{ value = 3.7 round = yes }
```

---

## if

Conditional statement that modifies the accumulator based on a condition.

Note that the limit is a math expression. It is considered true if the value is anything but 0.

### ### Example

---

```
{ value = x
```

```
  if = { limit = { value = x greater_than = 10 } add = 100 }
```

```
  else = { subtract = 1 } }
```

---

## every\_collection

Iterates over a named collection, applying statements to the accumulator for each element.

### ### Example

```

{ value = 0
  every_collection = {
    named_collection = my_collection
    add = element_value } }

```

## Army HQs: Player DLC Feature

You can now deploy generals as divisions, unlocking additional proximity bonuses, but risking your general being captured. This necessitates some tweaks and adds some new features. **If you have a general you don't want to be captured**, such as a country leader, you should put the argument

```
can_be_captured = no
```

In their character file. It goes right under the name block.

Units now support some new arguments:

```

deployed_leader_modifiers = {
  river_crossing_factor = 0.2
}

allow_in_army_hq = yes
allow_in_non_army_hq = no

required_dlc = { "Thunder at Our Gates" }

deployment_cost = 1

enable_ability = { deeper_dig_in }

```

And in history/general/, you have a new file to add: taog\_hq\_template.txt. This sets a generic HQ template, as well as any unique ones you might want! **You must have a hq template for every country in the game, or you will have CTDs!**

# Ship Captains: Player DLC Feature

The other headlining feature is ship captains, analogous to divisional commanders. Related to this, medals have been slightly tweaked. For example, what was formerly

```
unit_modifiers = {  
    army_org_factor = 0.05  
    army_strength_factor = 0.05  
}  
  
one_time_effect = {  
    add_divisional_commander_xp = 100  
}
```

Would now be

```
unit_modifiers = {  
    army_org_factor = 0.05  
    army_strength_factor = 0.05  
}  
  
ship_modifiers = {  
    navy_org_factor = 0.05  
    naval_critical_effect_factor = 0.05  
}  
  
officer_xp = 100
```

Officer\_xp is the new effect for xp!



Generally speaking, the following effects are used as equivalent in vanilla, with the exception of a few sign changes (to keep things buffs):

| Land Effect                   | Navy Effect                               |
|-------------------------------|-------------------------------------------|
| army_morale_factor            | naval_morale_factor                       |
| experience_loss_factor        | experience_gain_navy_unit_factor          |
| supply_consumption_factor     | naval_supply_consumption_factor           |
| recon_factor                  | naval_detection                           |
| max_dig_in_factor             | naval_mines_effect_reduction              |
| army_defence_factor           | naval_defense_factor                      |
| breakthrough_factor           | naval_critical_score_chance_factor        |
| army_attack_factor            | naval_damage_factor                       |
| army_org_factor               | navy_org_factor                           |
| army_fuel_consumption_factor  | navy_fuel_consumption_factor              |
| equipment_capture_factor      | naval_ship_recovery_chance                |
| recon_factor_while_entrenched | navy_visibility <i>or</i> spotting_chance |
| army_strength_factor          | naval_critical_effect_factor              |
| armor_factor                  | critical_receive_chance                   |

## Expanded Player Features

Here are the updated and new things that the player will care about that are not whole new categories.

## Special Forces New–Style Doctrines

Special forces now use the mastery system with tracks. This also allows for some new modding abilities:

```
allow_in_multiple_tracks = yes # Optional – if yes, this subdoctrine can be assigned to multiple tracks
*at the same time*. Default is no.

xor = { other_subdoctrine_a other_subdoctrine_b } # Optional – lists subdoctrines that cannot be
active simultaneously in a different track within the same folder. Replacing this subdoctrine with an
xor'd one on the *same* track is still allowed.
```

## New Critical Hit Modules

A huge amount of new modules for ships to crit have been added, for both hulls and modules! If you have unique or extended modules, consider extending them correspondingly.

## Expanded Modder Features

### More AI Navy Stuff

There is a new argument for taskforces:

```
keep_updated = yes
```

This feature isn't documented, but I assume it determines where the most modern ships go.

### Focus Inlay Progressbars

Focus inlay windows now support progressbars.

```
scripted_progressbars = { # list of progressbars that should fill up based on a variable

    icon_name = { # Name of the progressbar (must be an icon that is a subcomponent of
"gui_component_name" or a button, with a "progressbar" as its spriteType)
```

```
    progress = <variable_name>

}
```

## New Raid Options

We've got naval commando raids now. We've got some new modifiers:

```
    ai_min_success_chance = 0.1 # [Optional] Per-type override of
RAIDS_MIN_SUCCESS_FOR_LAUNCH (0.0-1.0).

    max_distance = 500 # [Optional] Hard cap on source-to-target path length, in universal distance
units (same space as the engine's path lengths). Sources further away than this cannot be used. AI
ignores this when unlimited_ai_range = yes.
```

As well, there is a whole new section for Custom success chance modifiers. Check the documentation!

## Overhauled Ability Modifiers

The way abilities work is entirely different now. The old syntax that you are familiar with:

```
one_time_effect = {

    add_temporary_buff_to_units = {

        combat_offense = 0.2

        combat_breakthrough = 0.25

        org_damage_multiplier = -1.0

        str_damage_multiplier = 0.6

        war_support_reduction_on_damage = 0.2
```

```

    cannot_retreat_while_attacking = 1.0

    days = 7

    tooltip = ABILITY_FORCE_ATTACK_TOOLTIP
}
}

```

And the new syntax that you now have to use:

```

unit_modifiers = {
    offence = 0.2
    breakthrough_factor = 0.25
    org_damage_multiplier = -1.0
    str_damage_multiplier = 0.6
    war_support_reduction_on_damage = 0.2
    cannot_retreat_while_attacking = 1.0
}

```

The old syntax **does not work anymore**. You must change it!

Additionally, a new argument is supported:

```

requires_deployed_hq = yes

```

## New Syntax for Dynamic Focus Icons

The old syntax that you are no doubt familiar with looked like this:

```

icon = {
    trigger = {
        has_government = democratic
    }
}

```

```

    value = GFX_focus_GER_german_hegemony_in_the_middle_east_demcoratic
}

icon = {

    trigger = {

        NOT = {

            has_government = democratic

        }

    }

    value = GFX_focus_GER_german_hegemony_in_the_middle_east_neutrality

}

```

That is gone. Now it's:

```

icon = {

    GFX_focus_GER_german_hegemony_in_the_middle_east_demcoratic = {

        has_government = democratic

    }

    GFX_focus_GER_german_hegemony_in_the_middle_east_neutrality = yes

}

```

The old syntax **does not work anymore**. You must change it!

## Synchronized Dynamic Tokens

Certain dynamic tokens, such as custom modifier\_definitions, have certain limitations. Which **could** cause OOS problems in multiplayer games. Now these tokens will, for most of them, give an error in the error log.

The error looks like this.

Token <token\_name> is a dynamic token, this can cause OOS depending on how it's used, please add it as a synchronized dynamic token to prevent OOS:es

These tokens can now be added to a text file in **Common/synchronized\_dynamic\_tokens** to remove the risk that they cause on OOS.

## New effects

### **release\_from\_captivity**

Character scope

Releases the scoped army leader from captivity. Has no effect if the leader is not captured.

#### ### Example

```
^^  
  
var:target_leader = { release_from_captivity = yes }  
  
^^
```

### **unlock\_subunit**

Country scope

Unlocks the specified subunit for the country.

#### ### Examples

```
^^  
  
GER = {  
    unlock_subunit = rangers_support  
}  
  
^^
```

### **release\_captured\_generals\_from**

### Country scope

Makes the scoped country release all captured generals from the specified country kept in the specified province.

If no province is specified, all generals are released.

#### ### Example

```
...  
  
release_captured_generals_from = {  
    target = GER  
    province = <province> (optional)  
}  
  
release_captured_generals_from = FROM  
...
```

### capture\_by

Make the Unit Leader in scope be captured by the specified country (needs to be a general).

Optionally specify the province where the leader is held captive.

#### ### Example

```
...  
  
capture_by = GER  
  
capture_by = FROM  
  
capture_by = {  
    country = GER  
    province = 1234  
}
```

```
}  
...  

```

## New triggers

### **can\_be\_captured**

Character scope

**can\_be\_captured = yes/no** – Checks if the current character can be captured while deployed in a division

### **is\_leader\_visible**

Country scope

**Checks if the leader is visible to its owner (is\_visible trigger in the leader's script)**

### **is\_in\_state**

Character scope

Checks if the character is in a specific state.

Works for deployed generals (HQ location), scientists (facility location including faction programs), and operatives (mission location).

### **### Example**

```
...  

```

**# Check if a general is in a specific state**

```
any_unit_leader = {
```

```
    is_in_state = 64 # Brandenburg
```

```
}
```

**# Check if any scientist is in the capital**



```

every_character = {
    limit = { is_active_scientist = yes }

    is_in_state = root.capital
}

```

### **has\_completed\_goal**

Country scope

Checks if the country was a member of a faction when a specific goal was completed.

Persists across faction disband/recreate, unlike has\_completed\_faction\_goal.

### ### Examples

```

TAG = {
    has_completed_goal = goal_id
}

```

### **debug\_math\_expr**

any scope

Evaluates a math expression and writes the numeric result to the tooltip.

Always returns true and is intended for debugging scripts.

### ### Parameters

– `Expression`: Any supported math expression.

### ### Example

```


```

```
debug_math_expr = { value = 120 add = 15 clamp = { min = 0 max = 100 } }  
^^^
```

### **has\_any\_general\_captured\_by**

Country scope

Checks if the scoped country has any unit leader captured by the target country

#### ### Examples

```
^^^  
FRA = {  
    has_any_general_captured_by = GER # This will be true if any of France's generals are captured by  
    Germany  
}  
^^
```

### **is\_general\_captured**

character scope

Checks if the scoped unit leader (general) is captured (by any country)

#### ### Examples

```
^^^  
my_character = {  
    is_general_captured = yes  
}  
^^^
```

### **has\_any\_captured\_general**

Country scope

Checks if any of the generals of the scoped country is captured by another country

### ### Examples

```

FRA = {

    thas\_any\_captured\_general = yes

}

```

### is\_promoted\_from\_unit

character scope

Checks if a unit leader was promoted from a unit (division or ship).

### ### Example

```

any\_unit\_leader = {

    is\_promoted\_from\_unit = yes

}

```

### check\_expr

any scope

Evaluates a math expression and returns true when the result is non-zero.

### ### Parameters

– `Expression`: Any supported math expression.

### ### Example

```

```
check_expr = { value = 2 add = 3 greater_than = 4 }  
...
```

## New modifiers

army\_non\_core\_attack\_factor  
cannot\_retreat\_while\_attacking  
army\_non\_core\_defence\_factor  
cannot\_retreat\_while\_defending  
casualty\_trickleback  
combat\_entrenchment  
org\_damage\_multiplier  
str\_damage\_multiplier  
war\_support\_reduction\_on\_damage

## New Variables

**days\_until\_next\_election**

days until the next election. -1 if never

**convoys\_destroyed**

total number of convoys destroyed by this country

**avg\_unit\_entrenchment\_ratio**

average entrenchment ratio of all units

## New Defines

**CAPTAIN\_EXPERIENCE\_ON\_SHIP\_MULT = 1.55,**

    -- Multiply ship experience gained by this, when applying to ship captain

**MAX\_CAPTAIN\_EXPERIENCE\_ON\_SHIP = 8000,**

    -- Max experience that can be gained on ship captains

**CAPTAIN\_EXPERIENCE\_ON\_SHIP\_PER\_MEDAL\_MULT = 0.1,**

    --Multiply ship captain experience gain by this \* number of ship medals on application

**MAX\_REGIMENTAL\_SUPPORT\_WIDTH = 5,**

    -- Max width of regimental supports in division designer.

**MAX\_REGIMENTAL\_SUPPORT\_HEIGHT = 1,**

    -- Max height of regimental supports in division designer.

**MAX\_HQ\_BATTALION\_WIDTH = 1,**

    -- Max width of regiments in division designer (Army HQ templates).

**MAX\_HQ\_BATTALION\_HEIGHT = 4,**

    -- Max height of regiments in division designer (Army HQ templates).

**MAX\_HQ\_SUPPORT\_WIDTH = 1,**

    -- Max width of support in division designer (Army HQ templates).

**MAX\_HQ\_SUPPORT\_HEIGHT = 4,**

    -- Max height of support in division designer (Army HQ templates).

**MAX\_HQ\_REGIMENTAL\_SUPPORT\_WIDTH = 0,**

    -- Max width of regimental supports in division designer (Army HQ templates).

**MAX\_HQ\_REGIMENTAL\_SUPPORT\_HEIGHT = 0,**

    -- Max height of regimental supports in division designer (Army HQ templates).

**REGIMENTAL\_SUPPORT\_REQUIRED\_BATTALIONS = { 3 },**

    -- For each regimental support row, how many battalions are required in the regiment to be able to place a support in that row.

    -- When the AI is deciding where to place battalions, it tries to place it in the position with the lowest number according to this grid.

**AI\_BATTALION\_BUILD\_ORDER = { 1, 6, 11, 16, 21,**

**2, 7, 12, 17, 22,**

**3, 8, 13, 18, 23,**

4, 9, 14, 19, 24,  
5, 10, 15, 20, 25 },

**REGIMENTAL\_SUPPORT\_SLOT\_COST\_MULTIPLIER = 0.5,**

-- Regimental support slot costs are scaled by this value compared to normal support slots

**REGIMENTAL\_SUPPORT\_SLOT\_COST\_MULTIPLIER = 0.5,**

-- Regimental support slot costs are scaled by this value compared to normal support slots

**UNIT\_LEADER\_MODIFIER\_COOLDOWN\_ON\_DEPLOY = 3,**

-- base time in days for a unit leader to be deployed while not already deployed.

Instantaneous if 0. Scaled by HQ template manpower.

**UNIT\_LEADER\_MODIFIER\_COOLDOWN\_ON\_DEPLOY\_MIN = 1,**

-- minimum cooldown in days for deploying a unit leader, even for very small HQ templates

**UNIT\_LEADER\_MODIFIER\_COOLDOWN\_ON\_REDEPLOY = 15,**

-- time in days for a unit leader to be redeployed to a new unit. Instantaneous if 0

**UNIT\_LEADER\_MODIFIER\_COOLDOWN\_ON\_WITHDRAW = 3,**

-- base time in days for withdrawing a deployed unit leader. Instantaneous if 0. Scaled by HQ template manpower.

**UNIT\_LEADER\_MODIFIER\_COOLDOWN\_ON\_WITHDRAW\_MIN = 1,**

-- minimum cooldown in days for withdrawing a unit leader, even for very small HQ templates

**UNIT\_LEADER\_MODIFIER\_COOLDOWN\_REFERENCE\_MANPOWER = 9600,**

-- reference manpower for deploy/withdraw cooldown scaling. At this value, delay equals the base.

**UNIT\_LEADER\_MODIFIER\_COOLDOWN\_MANPOWER\_EXPONENT = 1.5,**

-- exponent for manpower scaling (>1 = big divisions punished harder)

**FIGHTING\_STRENGTH\_DEATH\_THRESHOLD = 0.001,**

-- fighting strength below which divisions die from attrition

**FIGHTING\_STRENGTH\_HQ\_ALERT\_THRESHOLD = 0.2,**

-- fighting strength below which deployed leader have an alert glow on their portrait

COMMANDER\_ABILITY\_BASE\_RANGE = 20,

-- Base radius range of commander abilities

COMMS\_MAX\_DISTANCE = 4,

-- If N is >= the size of the below arrays, the last value will be considered repeated

PLANNING\_CAP\_COMMS\_SCALING = { 1.0, 0.95, 0.9, 0.85, 0.8 },

-- Value at index J is the scaling applied to planning cap when HQ is J provinces behind the  
frontline

PLANNING\_CAP\_NO\_HQ\_SCALING = 0.8,

-- Scaling applied to planning cap when there's no HQ (no leader or leader not deployed or  
not the same root order)

PLANNING\_SPEED\_COMMS\_SCALING = { 1.0, 0.95, 0.9, 0.85, 0.8 },

-- Same as PLANNING\_CAP\_COMMS\_SCALING but for planning speed

PLANNING\_SPEED\_NO\_HQ\_SCALING = 0.8,

-- Same as PLANNING\_CAP\_NO\_HQ\_SCALING but for planning speed

LEADER\_MOD\_COMMS\_SCALING = { 1.06, 1.04, 1.02, 1.01, 1.0 },

-- Same as PLANNING\_CAP\_COMMS\_SCALING but for leader modifiers

LEADER\_MOD\_NO\_HQ\_SCALING = 1.0,

-- Same as PLANNING\_CAP\_NO\_HQ\_SCALING but for leader modifiers

ABILITY\_COMMS\_SCALING = { 1.06, 1.04, 1.02, 1.01, 1.0 },

-- Same as PLANNING\_CAP\_COMMS\_SCALING but for active abilities

ABILITY\_NO\_HQ\_SCALING = 1.0,

-- Same as PLANNING\_CAP\_NO\_HQ\_SCALING but for active abilities

GENERAL\_PROXIMITY\_CLOSE = 1,

-- At the "close" proximity setting, the general should stay this many provinces behind the  
frontline

GENERAL\_PROXIMITY\_MEDIUM = 2,

-- At the "medium" proximity setting, the general should stay this many provinces behind the  
frontline

GENERAL\_PROXIMITY\_FAR = 3,

-- At the "far" proximity setting, the general should stay this many provinces behind the frontline

**GENERAL\_PROXIMITY\_DEFAULT = 1,**

-- The default proximity setting for a deployed general. This number should correspond to one of the values above

**GENERAL\_RANK\_TO\_ARMY\_HQ\_EXP\_LEVEL\_FACTOR = 0.5,**

-- The general's rank is multiplied by this factor (rounded up) to determine the spawned Army HQ division's experience level

**ARMY\_HQ\_REQUISITION\_MINIMUM\_REMAINING\_PERCENTAGE = 10,**

-- When deploying an Army HQ, divisions will not have their equipment or manpower requisitioned below this percentage of their target manpower or equipment

**PREFERRED\_PRISON\_VP = 5,**

-- When capturing a general, try to find a province with at least that much VP to imprison them. The effective prison VP can be lower than that, if the capturing country doesn't have any province with at least this amount of VP

**CAPITAL\_SHIP\_COMBAT\_RETREAT\_MULT = 0.5,**

-- Multiplier on combat retreat threshold for capital ships and carriers (they can take more punishment)

**NAVY\_REPAIR\_BASE\_SEARCH\_NON\_OPERATIONAL\_STR = 0.65,**

-- strength factor at or below which a fleet is considered non-operational by the AI, causing it to cancel the mission and send the fleet to repair

**NAVAL\_COMBAT\_MEDAL\_CHANCE = 24,**

-- 1/N chance that a ship gains a medal after participating in a battle

**NAVAL\_COMBAT\_MEDAL\_MIN\_DURATION = 48,**

-- Minimum hours the battle must have taken to gain a medal

**NAVAL\_COMBAT\_MEDAL\_LAST\_MEDAL\_LIMIT = 30,**

-- Minimum days before the ship can gain a new medal

**NAVAL\_COMBAT\_MEDAL\_ALLOW\_CONVOY = false,**

-- Do naval attacks on convoy count for medals



**AI\_TASKFORCE\_REQUIRED\_RESERVE\_RATIO = 0.2,**

-- Fraction of required TF optimal composition held in reserve for reinforcement (rounded up per type)

**LENDLEASE\_CONVOY\_OVERCOMMIT\_PENALTY\_INTERVAL = 3,**

-- For every N convoys the receiver is over capacity (including this lend-lease), apply -1 to AI acceptance score

**STOP\_TRAINING\_ACTIVE\_COMBAT\_RATIO = 0.10,**

-- ai halts all training when more than this share of its divisions are in active combat (reinforce instead)

**CONVOY\_DEFICIT\_BUILD\_BOOST\_MAX = 150,**

-- Cap on the convoy-deficit-driven boost added to the convoy build target. Prevents convoy panic from starving warship production when many convoys are lost at once.

**CONVOY\_DEFICIT\_BUILD\_BOOST\_MAX = 150,**

-- Cap on the convoy-deficit-driven boost added to the convoy build target. Prevents convoy panic from starving warship production when many convoys are lost at once.

**AI\_SHIP\_SWAP\_MIN\_DAMAGED\_SHIPS = 2,**

-- minimum number of damaged ships in a taskforce before AI considers swapping them to reserves

**AI\_SHIP\_SWAP\_DAMAGE\_THRESHOLD = 0.33,**

-- per-ship strength threshold below which the AI considers a capital/carrier damaged enough to swap to reserves

**AI\_REPAIR\_CANCEL\_MIN\_STRENGTH = 0.75,**

-- AI will pull non-reserve task forces out of repair and back on mission once they reach this strength

**AI\_SURFACE\_COMBAT\_FUEL\_RATIO = 0.70,**

-- fraction of navy fuel reserved for surface combat (patrols, strike forces, dominance)

**AI\_CONVOY\_DEFENSE\_FUEL\_RATIO = 0.15,**

-- fraction of navy fuel reserved for convoy escorts (remaining goes to convoy raiding)

**INVASION\_UNITS\_READY\_AT\_MIN\_PLAN = 0.75,**

-- units ready ratio required when plan value is at minimum threshold

**INVASION\_UNITS\_READY\_AT\_MAX\_PLAN = 0.25,**

-- units ready ratio required when plan value is very high (1.0+)

**AI\_NAVAL\_GOALS\_UPDATE\_FREQUENCY\_DAYS = 7;**

-- Regenerate naval AI objectives this often in days (affects performance)

**ARMY\_LEADER\_ASSIGN\_WITHDRAW\_DAYS = 7,**

-- General will withdraw from HQ after that many days without an order HQ can influence (frontline or naval invasion)

**ARMY\_LEADER\_MIN\_DIVISIONS\_FOR\_HQ = 10,**

-- Army must contain at least this many divisions before the AI will deploy an HQ unit for it

**MAX\_DEPLOYED\_ARMY\_HQS = 5,**

-- AI will not deploy more than this many army HQs at once

**MAX\_CAPTURED\_GENERALS\_TO\_STOP\_HQ\_DEPLOY = 3,**

-- AI stops deploying new army HQs once this many of its army leaders are captured

## **Bookmark Labels**

**Written by Omnissiah**

**The bookmark labels are a part of the recently added Bookmark filters system. To put this an easy way, a labels is a non-predetermined set of custom filters that you can add anywhere**

To use them, go to your bookmark file and do this:

```
bookmarks = {  
    bookmark = {  
        name = "AFTERMATH_NAME"  
        desc = "AFTERMATH_DESC"  
        date = 1007.1.1.12  
        picture = "GFX_bookmark_zebrica"  
        default_country = "HIP"  
        default = yes  
  
        filters = { label } ### If you use the vanilla ones, just put it after them
```

That is it. Your bookmark now support labels, and they will be displayed after the vanilla ones if you add any. How to add one? Here' s an example:

```
"TAG1" = {  
    history = ""  
    ideology = ...  
    label = { label_1 label_2 }  
    ideas = {  
  
    }  
    focuses = {  
    }  
}  
"TAG2" = {  
    history = ""  
    ideology = ...  
    label = { label_1 label_3 }  
    ideas = {  
  
    }
```

```

        focuses = {
            }
    }

```

This will add labels label\_1, label\_2 and label\_3 correspondingly. In case you select label\_1, both countries will show up. Selecting either of the remaining ones will only show either TAG1 or TAG2 correspondingly.

Some things you should know:

1. It is not mandatory to use vanilla filters. Removing both new\_content and continent from the filters list is absolutely possible (and I'd recommend you doing it)
  2. There is no hard limit on labels country can have
  3. You do not have to pre-define those anywhere. Basically, it is "add wherever you need it" . The label is added to the filters list if at least one country in the bookmark uses it.
  4. You cannot define the order of those labels. I am unsure how it is selected, but it seems that the order gets re-randomized once the game reloads or the labels list gets updated.
  5. The GUI behind it is counter-intuitive and there is not much you can do. The width of the list will make it that all elements (e.g labels) will be evenly spread to cover that width.
- To change the filters width, you need to change the filters OverlappingElementsBoxType in the frontendgamesetupview.gui file.
6. A localisation key for a label is the same key you used to define one. So if your country has a label\_1, you add it as a loc key. They are one line only, and you can only change the width of the text by adjusting the width of the corresponding button (country\_filter container in the same file).
  7. There is no tooltip for the filter, however, in order to add something similar you can abuse the one-line issue of them.

Adding a loc key like that:

"Communism\n\nThis country either starts with Communism as their main ideology or can switch to it as a part of its storyline"

Will make it so everything before the \n marker is displayed on the button and everything else if you hover over it.

8. No, unfortunately, you cannot make a vertical filters list, which is sad. You cannot add a second line either.